

Ian Ronk

Data Engineer · Urban-Dynamics Researcher

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Production data systems with a specialty in the data engineering behind research: Airflow, DuckDB and PostGIS pipelines that turn open spatial data into papers. Research seat: urban dynamics — gentrification, accessibility, housing markets. Languages: Dutch, English, German, Italian.

Experience

Head of Data — KR&A, Amsterdam (FinTech, European real estate) Jul 2025 – present
Data strategy, a team of 4–5, and AI initiatives for an alternative-data FinTech serving European real-estate funds and REITs. Led the aggregation pipeline homogenising listings across 13 EU countries and the Connectivity Score build — 13 servers, on budget, two weeks ahead of plan.

Medior Data Scientist — KR&A Jun 2022 – Jul 2025
ML models for property valuation and market analysis across 15+ European markets; work delivered for real-estate companies, pension funds and Eurostat. Monthly house-price index across 13 EU countries tested as a disaggregation indicator for Eurostat quarterly HPIs.

Junior Data Scientist — KR&A Oct 2021 – Jun 2022
Flood-occurrence prediction from alternative data (BSc-thesis project); data preparation, aggregation and visualisation.

Junior Full-Stack Developer — Exact (former SRXP), part-time Sep 2019 – Sep 2022
Enterprise expense-declaration software (EmberJS, PHP) under CI/CD and testing.

Research & open source

- **Calibrating Free Postcode Boundaries from OpenStreetMap** — preprint, arXiv August 2026. Seed-density-to-IoU calibration of an OSM-Voronoi pipeline (NL/DK references, transfer to BE, applied to Italy's 4,209 CAP polygons).
- **When Does Metro Infrastructure Capitalize into Property Prices?** — working paper. Phase-decomposed DiD, seven European cities, $n = 42,004$; wild-cluster-bootstrap inference.
- **MSc thesis:** agent-based model of gentrification (attractiveness/affordability), Amsterdam · Utrecht · Milan.
- **data-orchestration** (public repo): Airflow 3 pipelines behind the research, with custom operators (DuckDB SQL ingestion, SHA-256 snapshot guard), tests and CI — written up as a blog series.

Education

MSc Data Science & Business Analytics — Bocconi University 2023 – 2025
Thesis: an agent-based model of gentrification in European cities.

BSc Artificial Intelligence — University of Amsterdam 2019 – 2023

Stack

Python · SQL · PostGIS/Postgres · Airflow · DuckDB · Docker · GeoPandas · H3 · PyTorch · XGBoost · React/Next.js · LLM/RAG pipelines